RAHMAH ABDULKARIM

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EDUCATION

Rice University — **Houston, TX**

M.A. Computational & Applied Mathematics & Operations Research, GPA: 3.76 — *May 2026*

- **Relevant Coursework:** Modern Machine Learning, Bayesian Statistics, Stochastic Processes, Advanced Numerical Analysis, Natural Language Processing, High Performance Computing

Howard University — **Washington, DC**

B.S. Mathematics; Minors: Computer Science, Chemistry, Latin; GPA: 3.91 — *May 2024*

- **Relevant Coursework:** Data Structures and Algorithms, Real Analysis, Probability and Statistics II, Chemistry II & Lab, Organic Chemistry II & Lab, Physics I and II for Science and Engineering, Numerical Analysis

RESEARCH EXPERIENCE

Graduate Research Assistant — **Aug 2024–May 2026**

Rice University — *Houston, TX*

- Analyzed model failure modes by linking prediction errors to input characteristics and distributional variation, introducing data filtering and training adjustments to stabilize performance across varied conditions.
- Integrated clinical constraints into feature engineering and evaluation criteria, ensuring model outputs reflected physiologically meaningful behavior beyond purely optimized numerical metrics.
- Documented preprocessing, modeling, and evaluation workflows in reproducible, version-controlled code for collaboration with technical and clinical stakeholders.

REU Research Assistant, Google-Sponsored — **May 2023–Aug 2023**

Rice University — *Houston, TX*

- Designed data transformation pipelines mapping irregular ocean-bed velocity measurements into structured inputs suitable for learning, including spatial indexing and neighborhood construction to preserve geographic relationships.
- Assessed model behavior using predictive metrics, correlation, and residual analysis, flagging cases where learned patterns contradicted known physical trends and required data or model adjustments.
- Ran controlled comparisons between simpler baselines and graph-based models using MLflow, analyzing performance gains relative to added complexity to inform model selection.

Undergraduate Research Assistant — **Feb 2023–Dec 2023**

Howard University — *Washington, DC*

- Developed and compared predictive models for prediabetes risk, implementing a gradient-boosted tree approach and using statistical analysis to identify key drivers of early onset.
- Analyzed BMI–diabetes relationships across ethnicities using R, identifying subgroup-specific patterns and highlighting the need for more personalized risk modeling.
- Investigated interventions for insulin resistance using the Matsuda index, linking model outputs to clinically relevant measures and improving interpretability of predicted risk.

Undergraduate Research Assistant, Department of Mathematics — **Aug 2022–Dec 2022**

Howard University — *Washington, DC*

- Developed and implemented a supervised clustering algorithm in MATLAB for the automatic extraction of reference tissue regions in PET (Positron Emission Tomography) studies.
- Worked on improving the accuracy of PET imaging using $[^{11}C]$-(R)-PK11195 for assessing microglial activation in the human brain.

PROJECTS

DICOM-Native EchoNet

- Designed a DICOM-aware geometric reconstruction pipeline that reconnects EchoNet-Dynamic segmentation outputs to their source echocardiographic DICOM metadata, enabling physically calibrated left ventricular volume and ejection-fraction estimates from preprocessed video data and validating their agreement with CMR reference measurements.

RSNA Intracranial Aneurysm Detection (Kaggle)

- Constructed a full-stack 3D medical imaging ML pipeline for intracranial aneurysm detection and vascular-territory prediction, converting raw CTA DICOM series into standardized volumetric tensors and training a class-balanced multi-label 3D CNN with quantitative validation and Grad-CAM interpretability analysis.

Subsurface Hydrocarbon Detection

- Implemented a graph-based learning system to detect hydrocarbon-rich regions from ocean-bed velocity/signal measurements by engineering graph representations that preserve spatial relationships, training a Graph Neural Network to capture non-local dependencies missed by pointwise baselines, and pairing model development with statistical validation to quantify signal–target relationships.

Mathematical Modeling for Diabetes Prevention

- Built a predictive modeling workflow to identify insulin-resistance risk drivers using engineered clinical features (including BMI-linked metrics and the Matsuda Index), trained and validated models with statistical tests and subgroup analyses to assess robustness across populations, and translated results into interpretable risk signals for earlier screening and more personalized prevention strategies.

SKILLS & LANGUAGES

Statistical & ML Methods:	Bayesian modeling, statistical inference, uncertainty quantification, model evaluation, bias analysis, subgroup analysis, feature engineering
Programming & Tools:	Python (PyTorch, scikit-learn, pandas, NumPy, OpenCV), R, MATLAB, C++, SQL, Git, Docker, AWS, MLflow, Linux, Jupyter
Data Modalities:	Medical imaging (DICOM, CTA, echocardiography), structured clinical data, geospatial signal data
Instructional Tools:	Microsoft Office (Word, PowerPoint, Excel), Google Workspace, Jupyter Notebook

TEACHING EXPERIENCE

Independent Mathematics Tutor, Calculus **May 2025–Aug 2025**
Private *Houston, TX*

- Independently designed and delivered a full summer Calculus curriculum for junior high students, creating structured lesson plans from scratch tailored to each student's level and learning pace.
- Managed all aspects of instruction independently: lesson delivery, assignment creation, grading, and written feedback.
- Built foundational Calculus intuition in students well ahead of their grade level using concrete examples and incremental problem-solving.

Teaching Assistant, Department of Latin **Aug 2021–May 2024**
Howard University *Washington, DC*

- Designed and delivered lesson plans independently in the professor's absence; introduced varied instructional techniques to improve student retention and engagement.
- Proctored examinations, graded submissions, and provided detailed written feedback; held weekly office hours and recitation sessions.

Khan Academy Teaching Fellow **Sep 2023–May 2024**
Howard University *Washington, DC*

- Tutored 25 high school students in College Algebra, providing individualized one-on-one sessions aimed at reducing failure rates; graded assignments and provided detailed feedback.

Student Tutor, Algebra, Calculus, Pre-Calculus **Aug 2022–May 2023**
ACE Learning Support Services, Howard University *Washington, DC*

- Supported over 70 students using clear, structured problem-solving methods; identified gaps in understanding and adapted explanations to meet students at their level.

LEADERSHIP AND ACTIVITIES

- **First Year Representative, Muslim Students' Association:** Advocated for 15+ freshmen; planned 10+ events for a 400+ member community, Howard University (2021–22).
- **Secretary, CHANGO Afro-Latino Organization:** Maintained records, coordinated communications, and facilitated cultural programming, Howard University (2021–22).
- **Engineers Without Borders:** Led chapter communications and coordinated member updates, Howard University (2022–23).

VOLUNTEERING

Howard University Alternative Spring Break **March 2023**
New Orleans, Louisiana (60 hours)

- Tutored over 25 students in an after-school program.
- Led panels for high school students from underprivileged backgrounds on the importance of attending college.

- Participated in community service, including drainage cleanup in flood-prone areas.

PROFESSIONAL AFFILIATIONS & MEMBERSHIPS

Honor Societies:	Phi Beta Kappa, Eta Sigma Phi, Pi Mu Epsilon
Professional Organizations:	American Mathematical Society (AMS), Society of Women Engineers (SWE), Association for Computing Machinery (ACM)
Service Organizations:	Howard University Alternative Spring Break (HU ASB), Engineers Without Borders (EWB), Howard University Chapel, Muslim Student Association (MSA)

HONORS AND AWARDS

- **Capstone Scholar:** Awarded a $100,000 competitive scholarship based on demonstrated scholarly intellect, community service, and leadership.
- **Artishia & Frederick Jordan Scholar:** Awarded a $16,000 scholarship based on academic potential, good character, and community service.
- **Summa Cum Laude, National Latin Exam (2021):** Top 5% of exam takers nationwide.
- **Alain Locke Awardee:** Recognized for maintaining a perfect 4.0 GPA as an underclassman.
- Solveig Espelie Mathematics Scholar and Donald Cox Awardee.